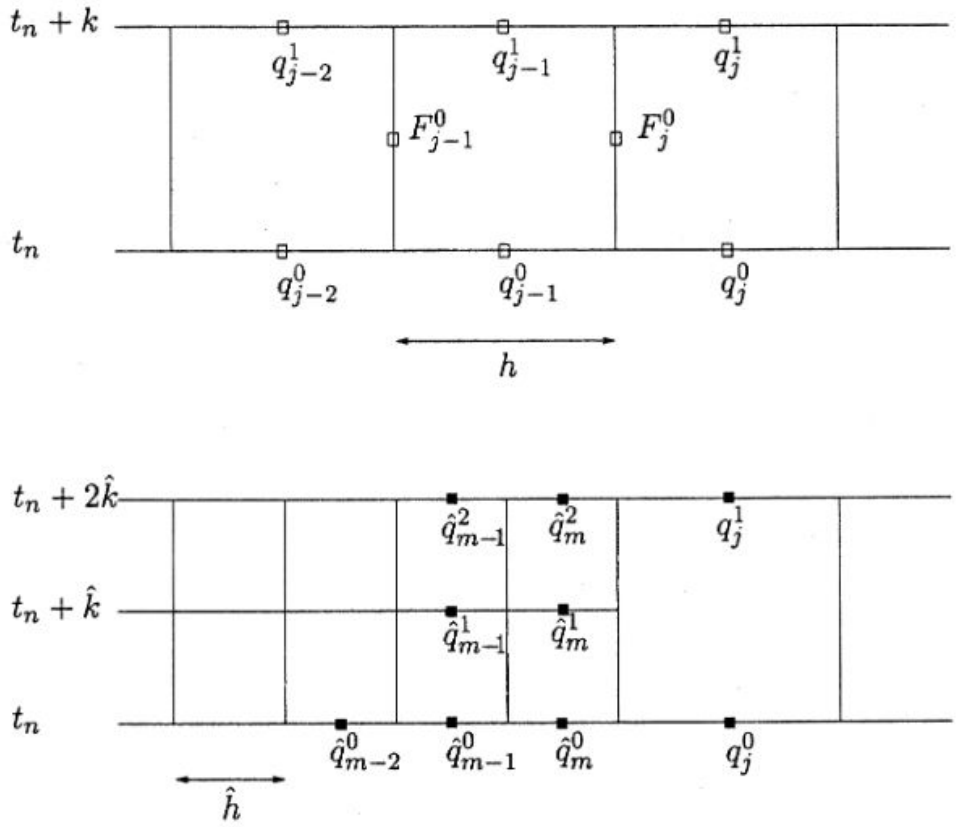
Adaptive Mesh Refinement

Principle of mesh refinement

Refinement

- a tagged cell of the original grid at level *L* is divided in R^L cells of the same size

- The time step of the original grid divided by R^L *is used to update the variables on the refined cells at level L*



FIGURE, Marsha J. Berger and Randall J. Leveque, Adaptive Mesh Refinement Using Wave-Propagation Algorithms For Hyperbolic Systems. Siam J Number Anal's, v 35, No 36, p 2298-2316, 1998.

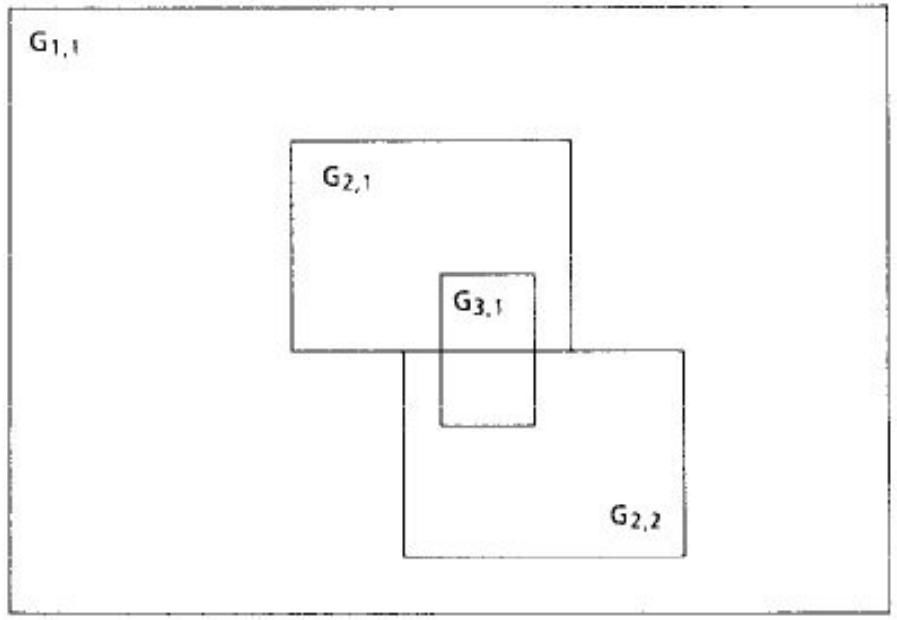
Principle of mesh refinement

Description of the grid

An AMR grid structure is usually a sequence of nested, uniform grids on which the PDE is discretized.

A level index is used to define the mesh size of the grid.

Each grid can be advanced independently, except for the boundary values.



FIGURE, Marsha J. Berger and P.Colella, Local Adaptive Mesh Refinement for Shock Hydrodynamics. Journal of Computational Physics 82, 64-84, 1989.

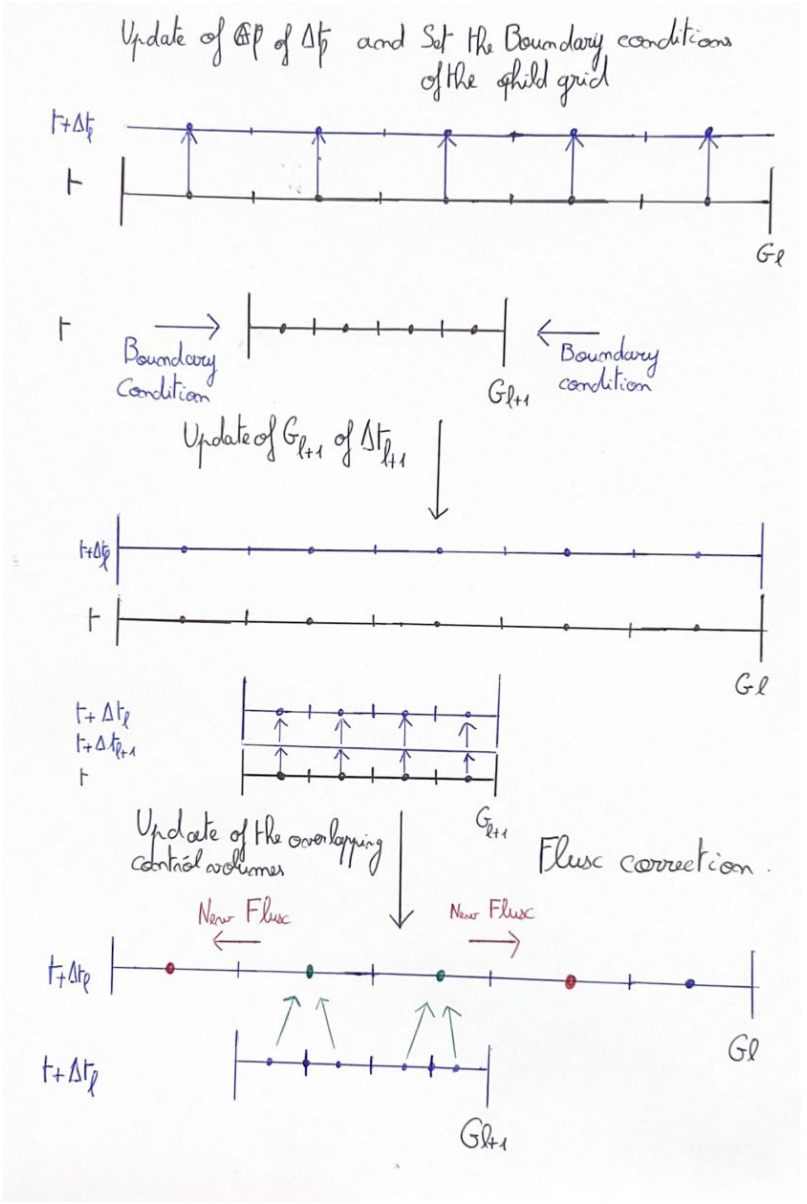
Principle of mesh refinement

Time step advancement algorithm

Algorithm 1 Update the grid hierarchy

DATA: Grid G at level l

- Update the grid G of Δt_l
- Set the boundary condition for each child grid of G
- Update each child grid $r = \frac{\Delta t_l}{\Delta t_{l+1}}$ times
- Update the values of the control volumes overlapping finer control volumes
- Correct the flux

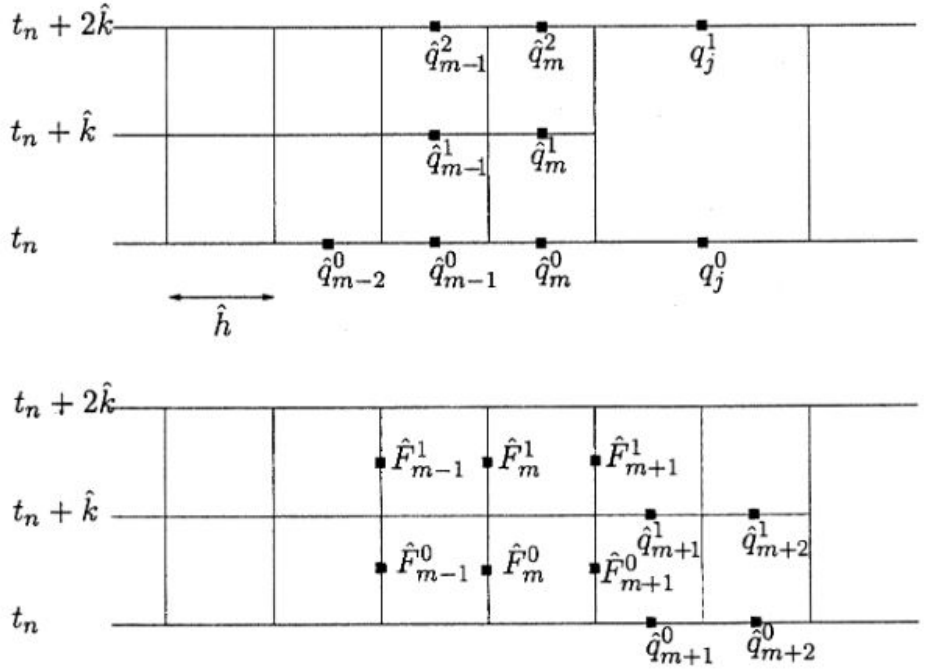


Principle of mesh refinement

Boundary conditions

- Boundary Conditions on a fine grid nested can be imposed using ghost cells.

- Ghost cells values are defined using space-time interpolation



FIGURE, Marsha J. Berger and Randall J. Leveque, Adaptive Mesh Refinement Using Wave-Propagation Algorithms For Hyperbolic Systems. Siam J Number Anal's, v 35, No 36, p 2298-2316, 1998.

Principle of mesh refinement

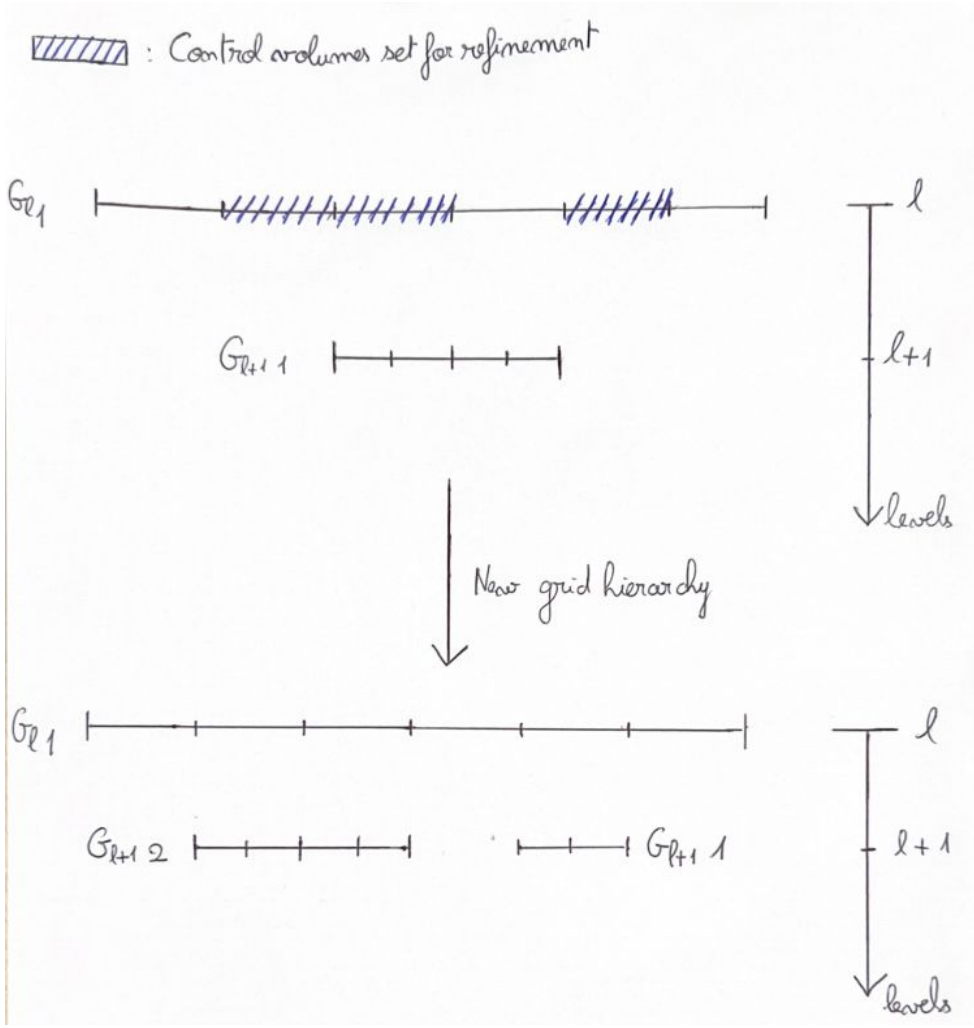
Adaptation of the grid hierarchy

Algorithm 2 Creation of the new grid hierarchy

DATA: Grid G at refinement level l
if The gird G does not have child grid or $l = l_{max} - 1$ **then**
- Apply the refinement criteria
- Restrict de child grids into the refinement areas
- Create new child grids and extend existing child grids to fit the refinement areas

  if $l < l_{max} - 1$ **then**
    Apply **Algorithm 2** to every child grid of G
  end if
else
- Apply **Algorithm 2** to every child grid of G
- Apply the refinement criteria
- Create new child grids and extend existing child grids to fit the refinement areas
- Apply **Algorithm 2** to every new child grid of G

end if



Refinement Criterion

Error estimator

Usually the refinement criteria is based on an error estimator. There are two types of error estimator:

A priori

- Evaluate where the most errors is made before the time step.
- Depends on the exact solution (usually the gradients of the solution)

A posteriori

- Evaluate where the most errors is made after the time step.
- Depends on the computing solutions.

Refinement Criterion

Example of error estimator

A posteriori estimator:

The error is estimated using the truncation error between to different level.

Richardson extrapolation is comparing the computed solutions using two time steps. One computed using the original grid and the other using a grid twice as coarser.

Refinement Criterion

Algorithm for Richardson extrapolation

Algorithm 3 Richardson extrapolation

DATA: Grid G at refinement level l, error tolerance ϵ, number of buffer zone N_b, time t

if l = 1 **then**

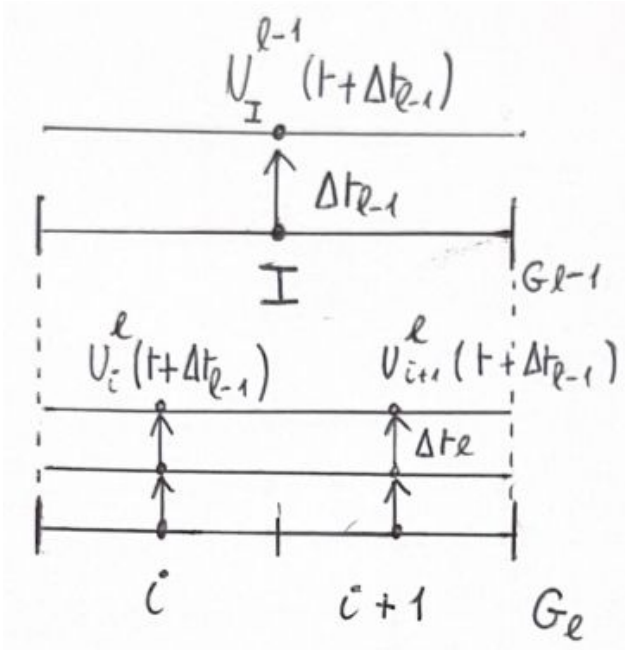
- Create the grid G_0 twice as coarse as G
- Update G_0 of Δt_0
- Update G of Δt_1 $r = \frac{\Delta t_0}{\Delta t_1}$ times
- Flag for refinement every couple of control volumes of G where $|U_I^0(t+\Delta t_0) - \frac{1}{2}(U_i^1(t+\Delta t_0) + U_{i+1}^1(t+\Delta t_0))| \geq \epsilon$

else

- Update G_{l-1} of Δt_{l-1}
- Update G of Δt_l $r = \frac{\Delta t_{l-1}}{\Delta t_l}$ times
- Flag for refinement every couple of control volumes of G where: $|U_I^{l-1}(t+\Delta t_{l-1}) - \frac{1}{2}(U_i^l(t+\Delta t_{l-1}) + U_{i+1}^l(t+\Delta t_{l-1}))| \geq \epsilon$

end if

- Flag for refinement every N_b control volumes next to an already flagged control volume



Refinement Criterion

Example of error estimator

A priori estimator:

In *Adaptive mesh refinement for two-phase slug flows with an a priori indicator* from M. Gourma, N. Jia and C; Thompson, a half grid of level *L* is refined if

$$\overline{\nabla U_{l,\pm}}^2 \geq \eta_l \quad \text{where} \quad \begin{cases} \overline{\nabla U_{l,-}}^2 = 2 \sum_{i=2}^{nl/2} (U_i - U_{i-1})^2 / nl \\ \overline{\nabla U_{l,+}}^2 = 2 \sum_{i=nl/2+1}^{nl} (U_i - U_{i-1})^2 / nl \end{cases} \qquad \eta_l = \left(\sum_{k=1}^{l} \overline{\nabla U_{k,-}}^2 + \overline{\nabla U_{k,+}}^2 \right)^{\frac{1}{2}} \Big/ \sqrt{2Nl}$$

FIGURE,M. Gourma, N. Jia, C. Thompson, Adaptive Mesh Refinement for Two-Phase Flow with an a priori indicator.International Journal of Multiphase Flow,2013.

Refinement Criterion

Example of error estimator

This criteria was not sufficient to ensure a good treatment of the slug distribution and slug frequency, in their case.
They added a *Kelvin-Helmholtz* instability detector, using *TD-KH condition.*

$$u_g^2 \geqslant \left(1 - \frac{h_l}{D}\right)^2 \frac{(\rho_l - \rho_g) \cdot g \cos\beta}{\rho_g} \cdot \frac{\alpha_g}{d\alpha_l / dh_l}$$

FIGURE,M. Gourma, N. Jia, C. Thompson, Adaptive Mesh Refinement for Two-Phase Flow with an a priori indicator.International Journal of Multiphase Flow,2013.

Refinement Criterion

Algorithm for the a priori estimator

Algorithm 4 Apriori estimator

DATA: Grid G at level l
if $l < l_{max}$ **then**
  Compute $\overline{\nabla U_{G,-}}^2$ and $\overline{\nabla U_{G,-}}^2$
  if $\overline{\nabla U_{G,\pm}}^2 \geq \eta_l$ **or** a control volume of $G_\pm$ succeed the Kelvin-Helmholtz instability test **then**
    Flag for refinement the half grid $G_\pm$
  end if
  Apply **Algorithm 4** to G_+ and G_-
end if

Refinement Criterion

Example of error estimator

Example:

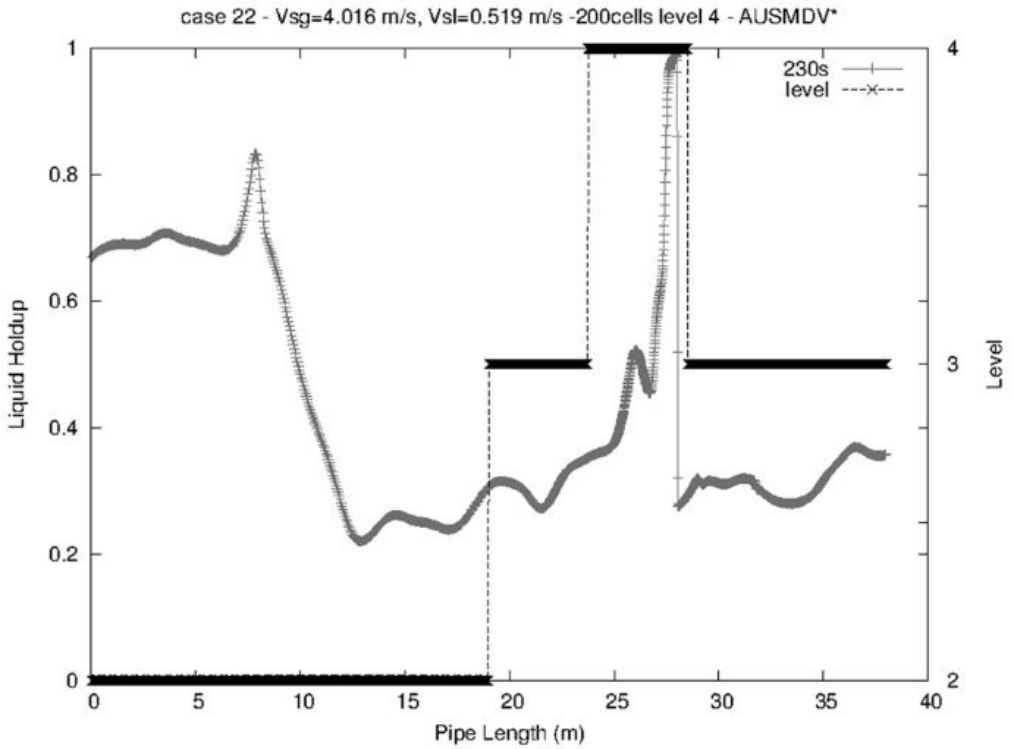


Fig. 15. Liquid distribution and grid level along the pipe at $t = 230$ s.

FIGURE,M. Gourma, N. Jia, C. Thompson, Adaptive Mesh Refinement for Two-Phase Flow with an a priori indicator.International Journal of Multiphase Flow,2013.